doubled and the frequency is reduced to one-fourth. Calculate now the new intensity of sound at the same point ?.

Answer: $I_{new} \propto \frac{1}{4} I_{old}$.

7. Consider two organ pipes of same length in which one organ pipe is closed and another organ pipe is open. If the fundamental frequency of closed pipe is 250 Hz. Calculate the fundamental frequency of the open pipe.

Answer: 500Hz

8. A police in a siren car moving with a velocity 20 ms^{-1} chases a thief who is moving in a car with a velocity $v_0 ms^{-1}$. The police car sounds at frequency 300Hz, and both of them move towards a stationary siren of frequency 400Hz. Calculate the speed in which thief is moving. (Assume the thief does not observe any beat)

Answer: $v_{thief} = 10\ m\ s^{-1}$

9. Consider the following function

(a) $y = x^2 + 2\ \alpha\ t\ x$

(b) $y = (x + vt)^2$

which among the above function can be characterized as a wave ?.

Answer: (a) function is not describing wave
(b) satisfies wave equation.

V. Conceptual Questions

1. Why is it that transverse waves cannot be produced in a gas?. Can the transverse waves can be produced in solids and liquids?
2. Why is the roar of our national animal different from the sound of a mosquito?
3. A sound source and listener are both stationary and a strong wind is blowing. Is there a Doppler effect?
4. In an empty room why is it that a tone sounds louder than in the room having things like furniture etc.
5. How do animals sense impending danger of hurricane?
6. Is it possible to realize whether a vessel kept under the tap is about to fill with water?

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UNIT-11(XI-Physics_Vol-2)_14-08-2020.indd 279 17-08-2020 15:32:08